

Final Project

Phase III - Dashboard Implementation

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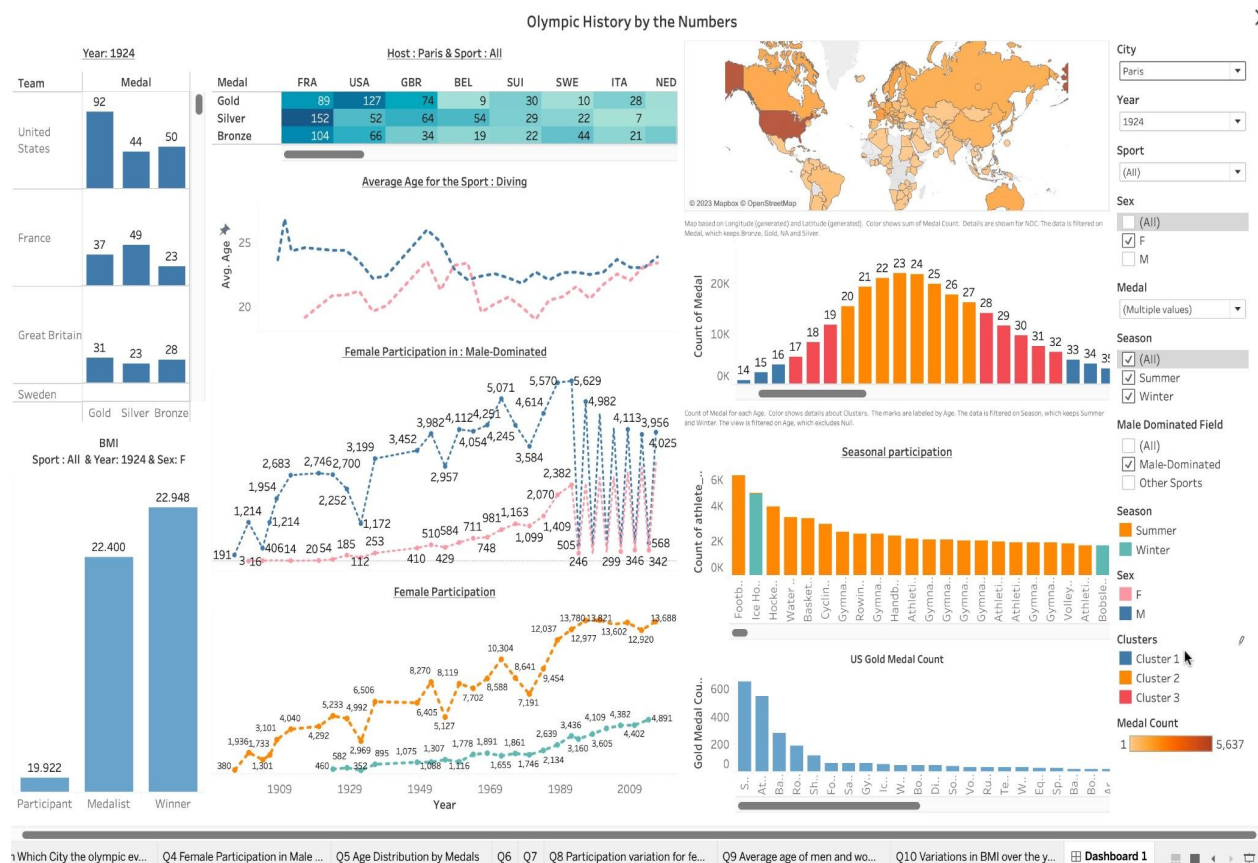
IFT 533: Data Visualization & Reporting for IT

Professor: Asmaa Elbadrawy

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Section 1: The Dashboard

Dashboards act as dynamic visual compilations, seamlessly integrating diverse datasets through tables, charts, and graphs for a lucid and comprehensive presentation. The primary goal is to provide a thorough overview of vital data, fostering simplicity in managing, assessing, and analyzing information. Tableau, a robust data visualization tool, enhances dashboard effectiveness with a user-friendly interface, real-time updates, and dynamic visualizations. It offers high customization, allowing users to tailor visuals and interactivity, and integrates parameters for dynamic manipulation. Tableau empowers users to create insightful dashboards, transcending traditional data representation and accommodating various industries and use cases. Overall, Tableau dashboards simplify data management and elevate the data-driven decision-making process.



Final Dashboard: Olympic History by the Numbers

Embark on a 120-year expedition, traversing the annals of Olympic history from 1896 to 2016, offering a detailed exploration of athletic events, their outcomes, and the evolving dynamics of global sportsmanship. This expansive dataset not only delves into the physical attributes of athletes, such as height, weight, and medal achievements by this dashboard.

Dynamically visualizing the seasons of dedication, stacked bar graphs depict the ebb and flow of athlete participation across Olympic seasons in this dashboard by providing athletes with insights into trends and sports analysts with strategic planning tools. Transitioning seamlessly, a multi-series bar chart vividly portrays the historical medal achievements of countries, offering athletes contextual understanding and sports analysts strategic insights.

This Dashboard is Delving into the intricate details of medals earned by teams in different host cities, the visualization unfolds insights valuable for athletes and coaches, guiding them in anticipating potential challenges and opportunities associated with specific locations. Breaking gender barriers in sports, the dataset explores the evolving landscape of female participation in traditionally male-dominated sports, inspiring athletes and providing strategic insights for sports analysts.

Shifting focus to demographics, an age distribution chart offers nuanced insights into the age dynamics of gold medalists, shaping understanding for athletes and coaches on age-related trends and peak performance. Beyond individual achievements, the global distribution of Olympic medals unfolds on a world map, fostering a global perspective for athletes and providing sports analysts with insights into international competition dynamics.

Sport-specific gold counts chart the sum of gold medals for each sport, offering athletes context for training goals and providing sports analysts with insights into national excellence. Comparative charts showcase the growth of female participation in Olympic sports, motivating athletes and guiding gender balance initiatives for sports analysts.

Beyond the numerical, a line chart thoughtfully explores how the average age of participants has changed over time, informing athletes and coaches about age-related trends and optimal performance. Illustrating changes in the average BMI of participants over the years, the chart provides insights for athletes and coaches into evolving physical standards and tailored training approaches.

In essence, this dashboard is a rich tapestry of Olympic history, offering a valuable tool for athletes, sports analysts, coaches, and enthusiasts alike. Providing nuanced insights, it seamlessly weaves historical data with future aspirations, making the journey through the Olympic legacy an inspiring and enlightening experience.

Maximizing Data-Ink Ratio in the Dashboard:

- **Optimized Label Placement:** The majority of charts employed direct numerical labels adjacent to bars or points, eliminating the need for y-axis headers. This strategic decision reduces unnecessary visual elements, enhancing the overall clarity of the visualization.
- **Streamlined Gridlines:** To mitigate visual clutter and optimize data density, gridlines were intentionally omitted from most charts, contributing to a cleaner and more focused presentation of information.
- **Fine-Tuned Width of Elements:** In pursuit of increased data-ink ratio, the width of bars and lines across charts has been meticulously minimized. Additionally, the size of bubbles and pie chart components has been scaled down, prioritizing the visual representation of relevant data.
- **Elimination of Redundant Visualizations:** The dashboard avoids redundancy in visual elements, ensuring that each chart serves a unique and purposeful role in conveying information.
- **Strategic Color Usage:** The application of color is judiciously employed to enhance data legibility without introducing unnecessary complexity. By minimizing color usage, the dashboard promotes a cohesive and easily interpretable visual experience for users.

Section 2: The Dataset

Description

Covering a span of 120 years from 1896 to 2016, this extensive dataset provides a thorough examination of Olympic history, encompassing various athletic events and their outcomes. Beyond revealing athletes' physical attributes like height, weight, and medal achievements, the dataset explores the diverse spectrum of sports featured in the Olympics. From the dynamic pace of hockey to the strategic finesse of basketball and the agility of badminton, each sport adds vibrancy to the tapestry of Olympic history.

Beyond individual achievements, the dataset unfolds the narrative of Olympic events, outlining the locations, years, and seasons that have hosted the pursuit of athletic excellence. Additionally, it illuminates the array of sports embraced by the Olympics, showcasing the diversity of competitions, ranging from track and field to team sports like hockey and basketball. This all-encompassing dataset beckons exploration into the triumphs, challenges, and enduring spirit of athleticism, offering a holistic perspective on the evolution of the Olympic Games over the years.

Rows: 217K

Columns : 15

This section explains the characteristic attributes of the data gathered.

Columns	Data Type Description	Domain & Possible Use Case
ID	This column contains individual identifiable participant ID numbers for all athletes involved in the	The ID column can be used as identifiers similar to the name which can help track an athlete of interest in multiple

	Olympics game. It is a nominal data	visualizations
Name	This column provides participant names. Data type is nominal	The name column can help to find the specific athlete
Sex	This column provides categorical data that groups participants by biological sex	The sex column provides information that can be used to compare different genders and takes values of Male or Female.
Age	This column provides information about the age of the participants. The data type is ratio	This column provides information that enables comparison between different age groups. It contains integers between 9 - 100 measured in years
Height	This column provides information on the height of participating athletes. The data class is ratio	Height is a ratio attribute measured in centimeters. This values could be analyzed to understand how physical attributes influence performance
Weight	This column contains information on the weight of players.	Weight is measured in kilograms and could be combined with other attributes to understand how it may influence performance
Team	Team is a nominal attribute that contains	This could also be used to compare different country

	information about the country athlete is representing	performances in the Olympics
NOC	NOC is the code used to represent the country of an athlete. The data type is nominal	This information could be used for axis labels, or as prettentive attributes in the visualization
Games	It is a nominal data which contains information the describes the year and season of the olympics tournament the athlete participated in	This column combines information between different years and seasons.
Year	This is an ordinal data that shows the year in which the event happened	This is an ordinal attribute that shows calendar years
Season	The season is a categorical dataset that tells when the event took place	Season contains categorical labels of Winter or Summer. It could be used to compare how performance records of different teams in different seasons.
City	This column contains nominal data about the hosting city of the event	Nominal label of describing city in which the event took place.
Sport	This column contains categorical attributes of the type of sport the	These attribute denotes the sport name such as curling, Ice hokey etc,. in text format.

	athlete participated in.	
Event	Contains categorical information describing the specific type of sport	This column could be used as filter to allow users to see information about name of the sport event they are participating
Medal	Ordinal Data type which contains information that describes the outcome from the game participation.	This column takes the values of gold, silver, bronze or NA(Not Achieved).

Data Preprocessing:

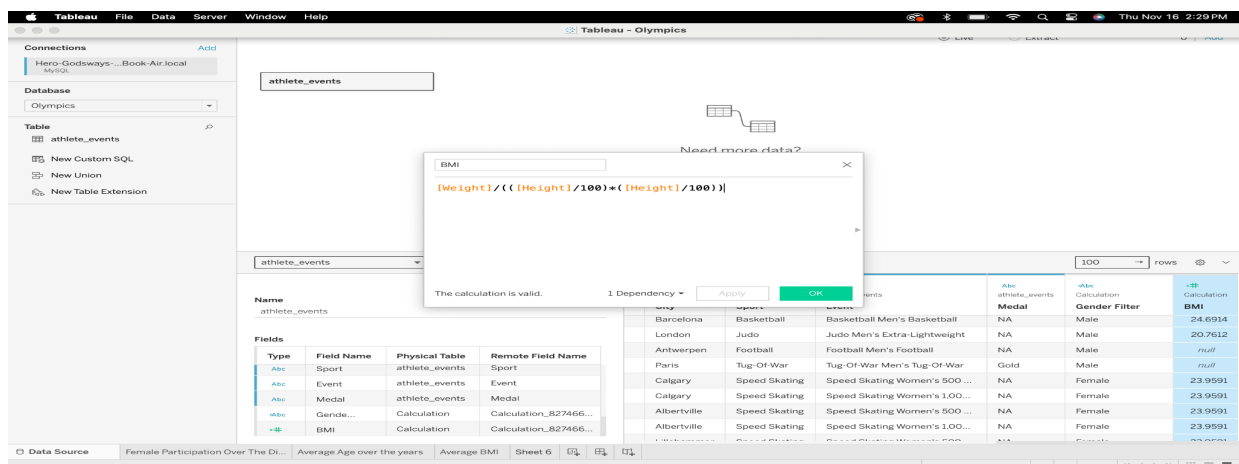
The selected dataset underwent necessary pre-processing to make it suitable for visualization, and Tableau and Python were employed as the tool for this data preparation.

Creating BMI Column using Calculated Field:

To compute the BMI of participating athletes, data preprocessing was computed using Tableau. The data from individual athlete parameters such as weight and height were combined using the formula $\text{Weight}/(\text{height})^2$.

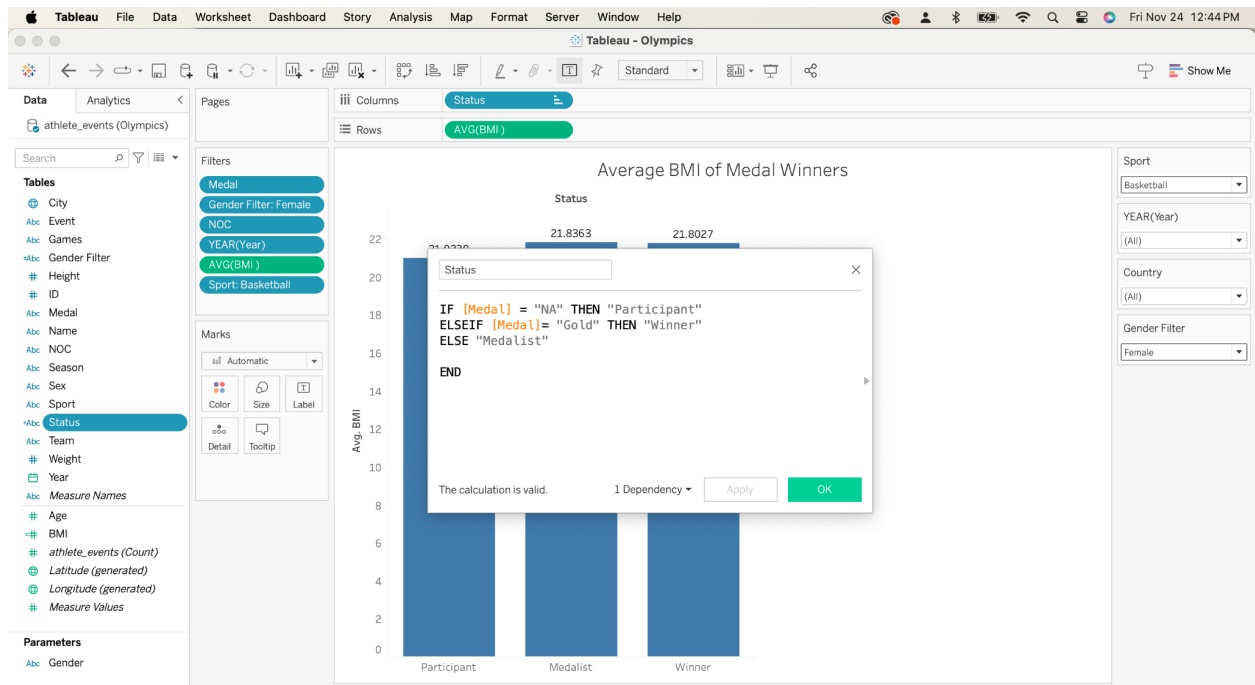
Preprocessing steps

1. Convert the height from centimeters to meters by dividing by 100. ($\text{height}/100$).
2. Divide the recorded athlete weight by the square of the height (kg/m^2)
3. Preprocessing Calculated Field



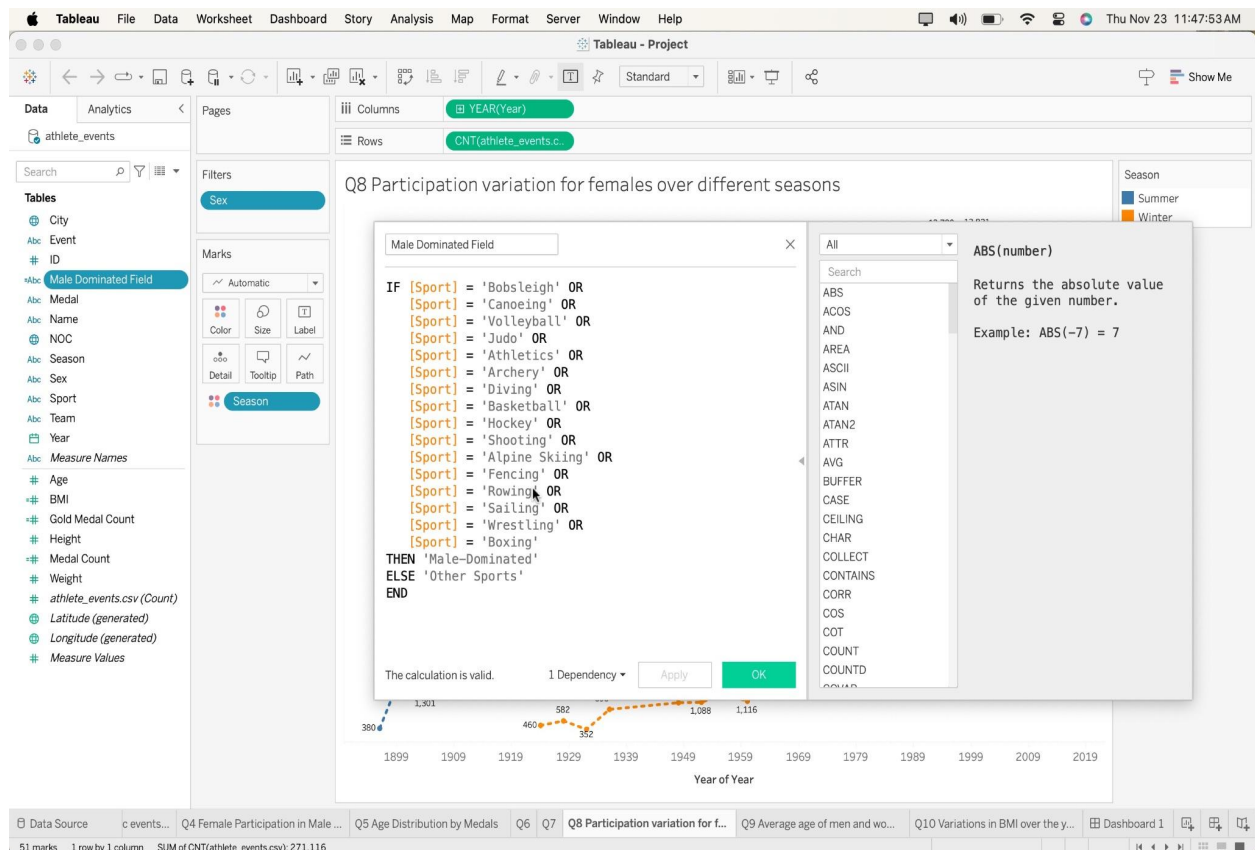
Creating a status label for medalist

To differentiate the average BMI of winners from other participants and bronze or silver medalists, a status calculated field was added to the columns. The formula used is shown in the screenshot below.



Creating a column of Male Dominated Sports using a Calculated Field

To compute the sports that have been male dominated in the history of the olympics, data preprocessing was done to streamline male prevalent sports from other sports, in order to determine the increase in female participation over the years.



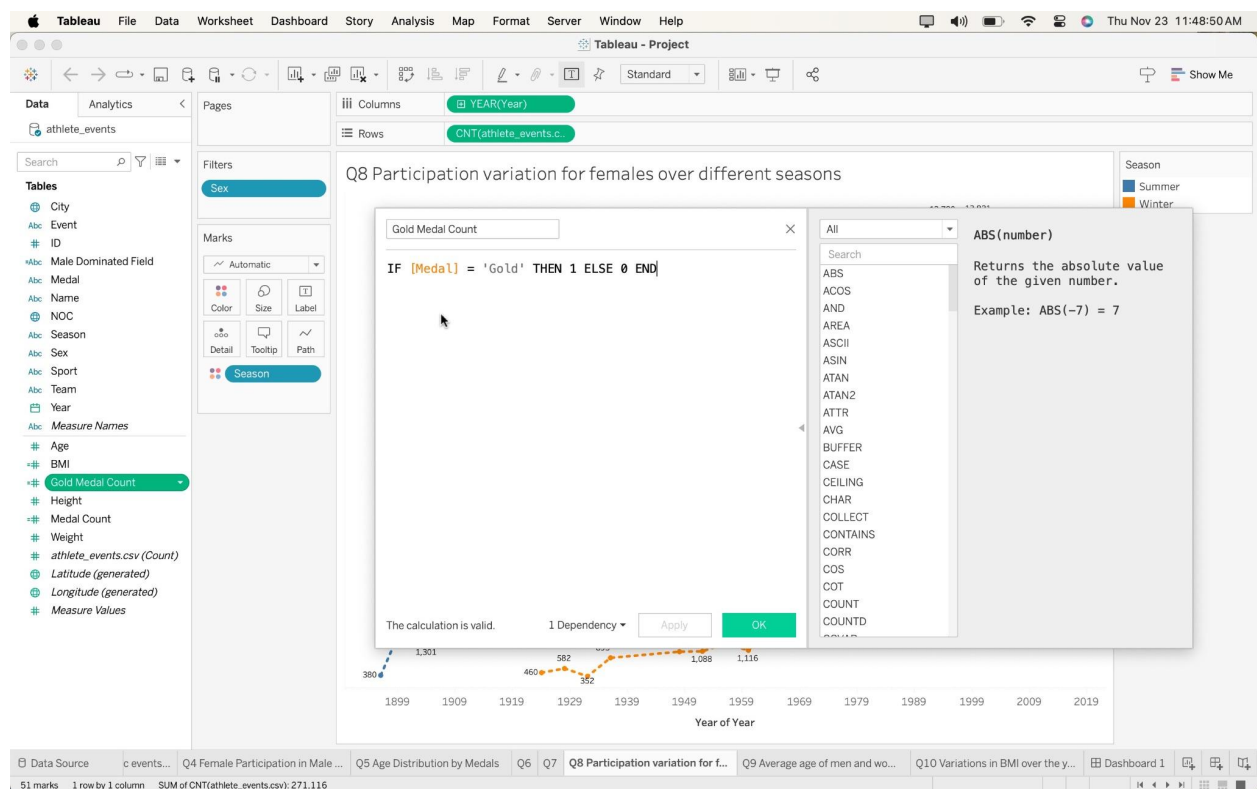
The calculated field makes all the olympic sports into two categories “male-dominated” and “other sports”, thereby allowing us to differentiate and analyze the trend in either growing or declining participation for either gender.

Creating a column of Gold Medal Count using a Calculated Field

We wanted to show the least number of gold medals won by the United States in each discipline in order to determine the sports in which they have been underperforming. This would require some pre-processing to calculate the number of gold medals for each sport and further filter it by country/region.

Gold Medal Count

```
IF [Medal] = 'Gold' THEN 1 ELSE 0 END
```



Creating a column for Medal Count using Calculated Field

This calculated field is a simple IF statement that checks whether the "Medal" field is not null. If a row in the dataset has a non-null value in the "Medal" column, it assigns a value of 1; otherwise, it assigns a value of 0. We do this to sum, calculate/count the number of medals in order to see the distribution of medals globally

Medal Count

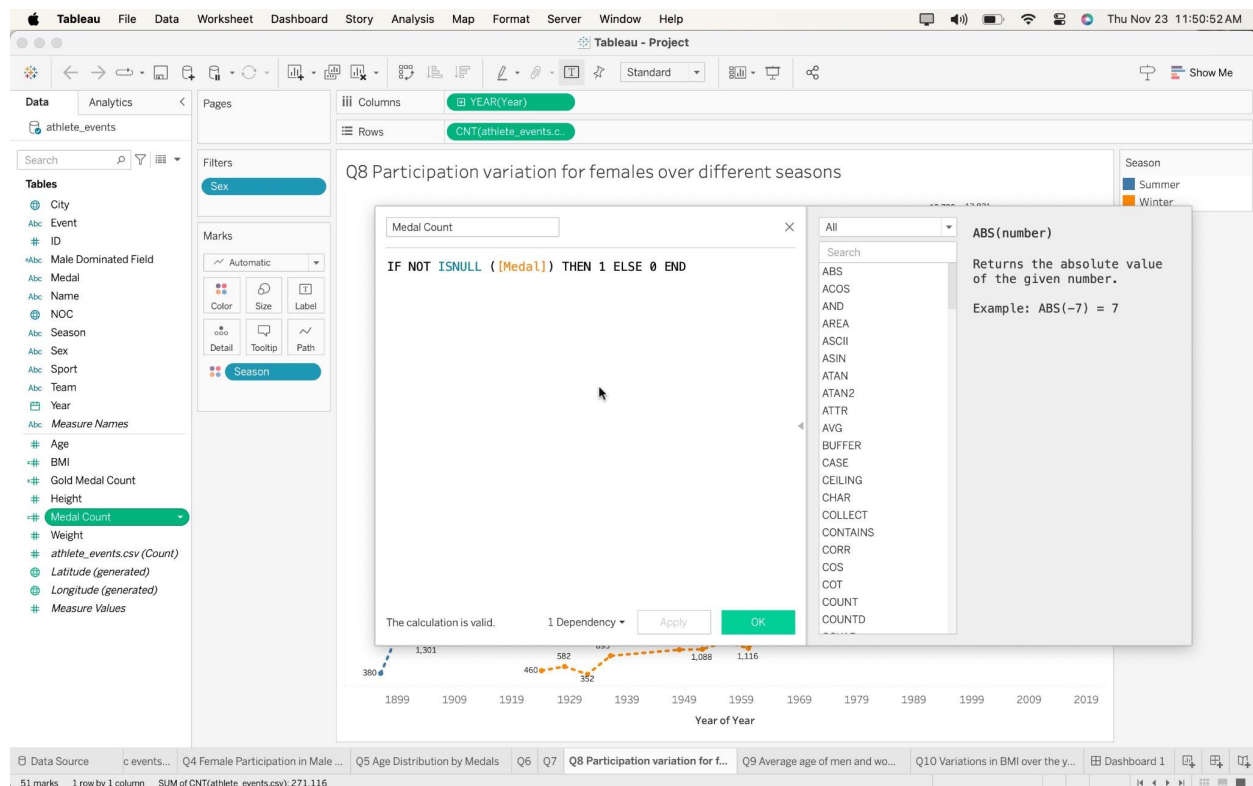
```
IF NOT ISNULL([Medal]) THEN 1 ELSE 0 END
```

The calculation is valid.

1 Dependency ▾

Apply

OK



Section 3: Dashboard Users

Athletes: Athletes can utilize the dashboard to benchmark their performances against historical data, gaining insights into successful strategies and areas for improvement. They can analyze the performances of past Olympic medalists, understand the competitive landscape, and make informed decisions about training priorities. The dashboard provides a valuable resource for athletes to set realistic goals, track progress, and draw inspiration from the achievements of their predecessors. Additionally, athletes can use the dashboard to explore the history of their respective sports, learn about the experiences of iconic athletes, and connect with the rich heritage of the Olympic Games.

Athlete Coaches: Coaches can leverage athlete-specific information such as height, weight, and performance metrics to tailor training programs. They can identify successful training regimens, evaluate the impact of specific physical attributes, and make data-driven decisions to enhance athletes' preparation for Olympic competitions.

Sports Analysts: Sports analysts can delve into the dashboard to perform statistical analyses, identify performance trends, and generate insights into the strategies employed by athletes and teams over the course of Olympic history. They can examine medal counts, winning patterns, and correlations between athletes' physical attributes and success.

Event Planners: Event planners can use historical data on Olympic events, including venue choices, logistical details, and audience engagement, to optimize the planning and execution of future Olympic Games. Insights from past events can inform decisions on venue selection, scheduling, and infrastructure development.

Section 4: Questions

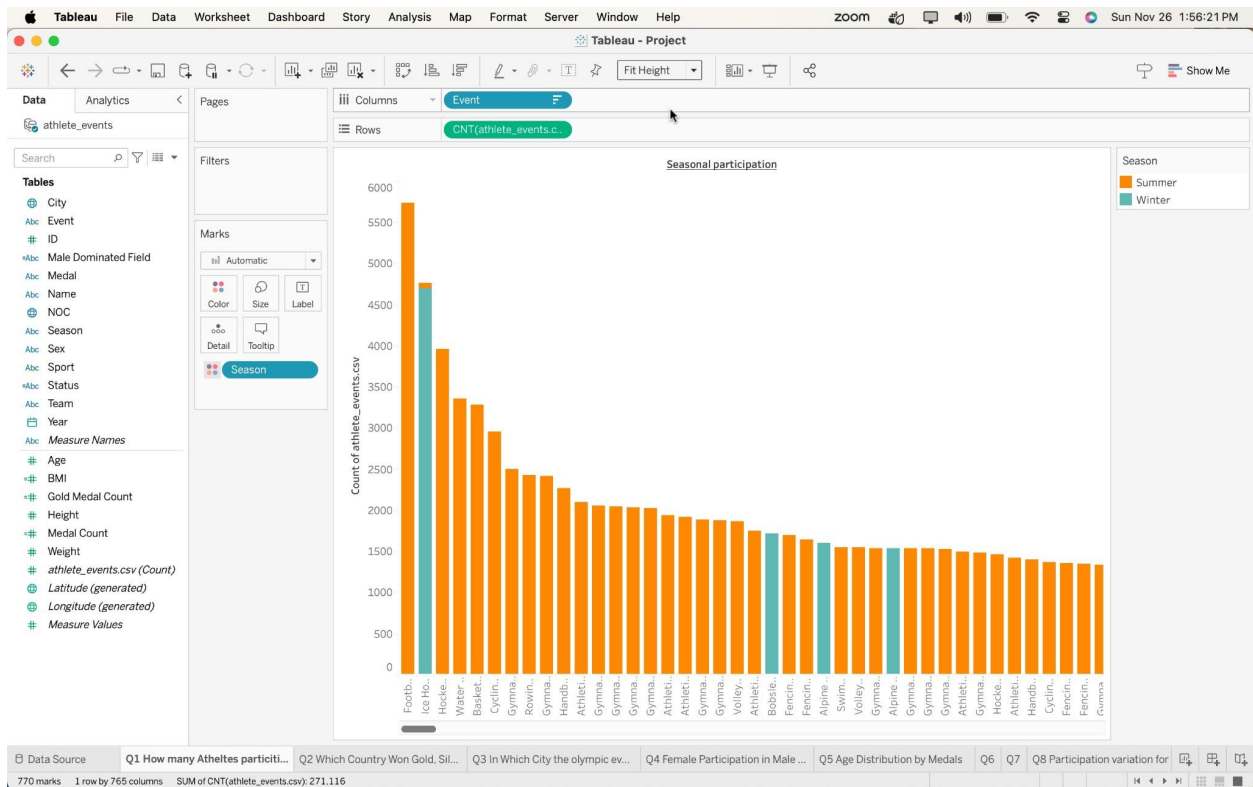
1. How many Athletes participated in the respective event based on season(Summer or Winter)
2. In the past 120 years of Olympic events, what is the count of gold, silver, bronze for each country in different years and overall years?
3. For each host country, what is the count of different medals achieved by the different teams.
4. Has there been variations in female participation over the years particularly in often male dominated sports.
5. What is the distribution of the age of gold medalists in either the summer or winter seasons?
6. What is the global distribution of medals across regions?
7. In the individual sports categories, where has the United States struggled to achieve medals?
8. What is the variations for female participation for the different seasons over the years
9. For selected teams(countries), what is the average age of women and men participating in the Olympics over the years
10. For the different sports, how has the average BMI of winners changed over the past years compared to non-winning participants?

Section 5: Plots

1. How many Athletes participated in the respective event based on season(Summer or Winter)

This visualization can represent how many athletes have participated in different seasons. In such a way, we plotted a stacked bar graph which represents the seasons of each bar with two different colors. To make this visualization possible, we considered **Event** attribute and **CNT(Athlete_Events)** which is the count of total athletes in the dataset. Finally Seasons for getting different colors while stacked. The Pre-Attentive attributes that we can identify are the length and color.

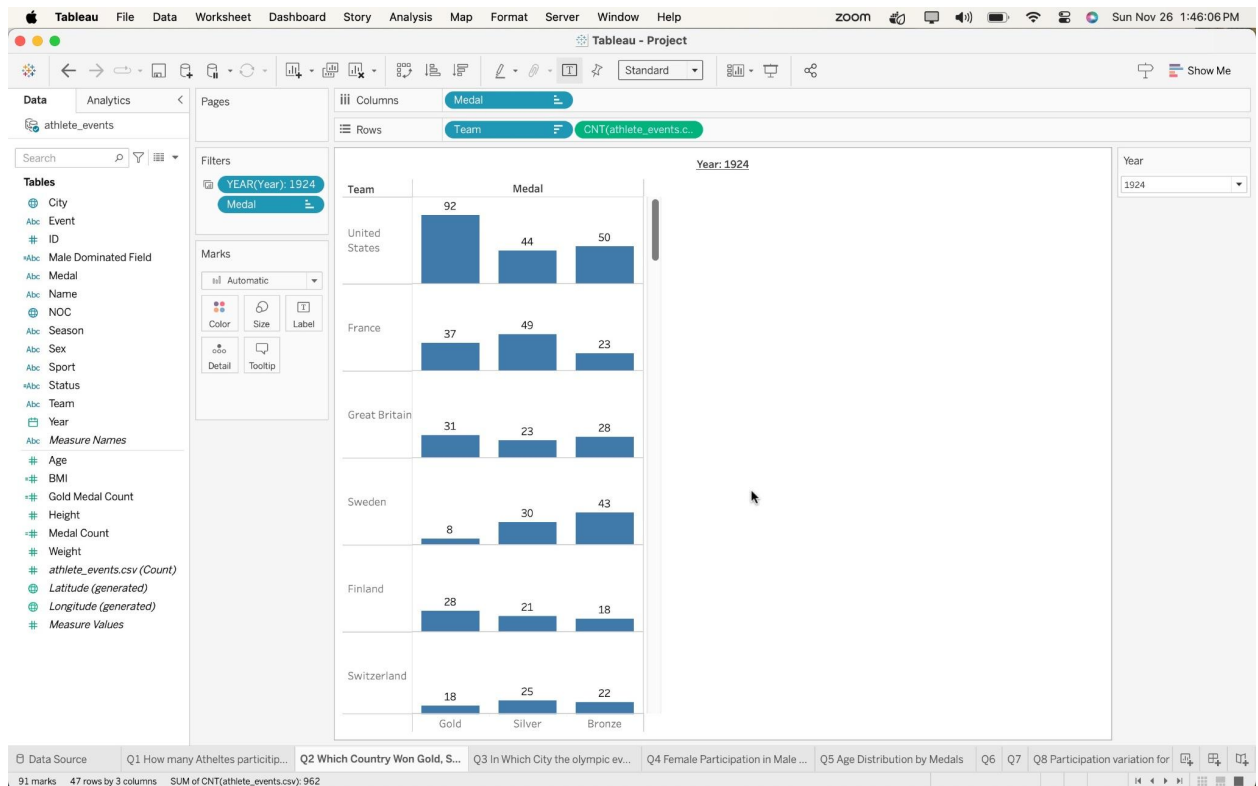
- **Length:** This defines the total number of athletes participating in each event.
- **Color:** Different seasons in bars are represented with different colors. Here, Blue refers to Summer and Orange represents Winter.

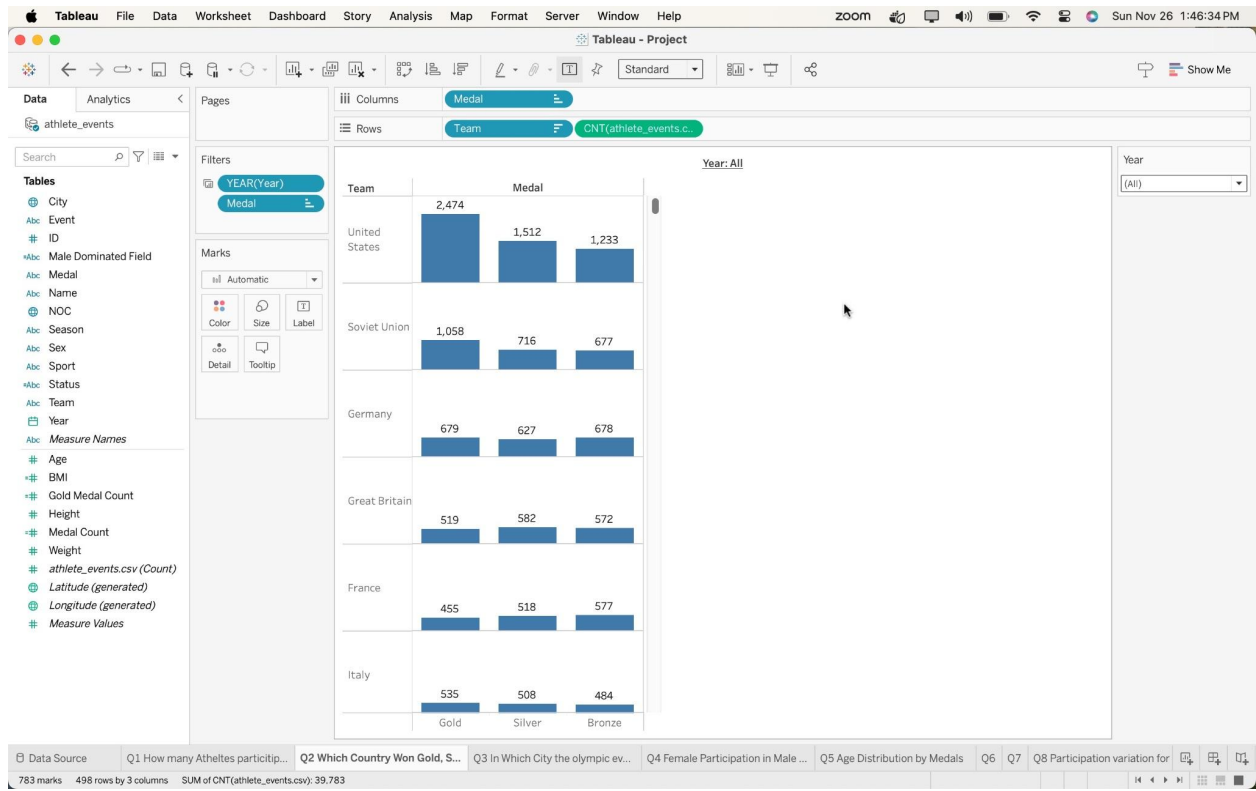


2. In the past 120 years of Olympic events, what is the count of gold, silver, bronze for each country in different years and overall years?

This visualization shows the country medals earned in different years. By this way we can visualize how each country got more medals in the different years. To make this visualization possible, I used a multi series bar chart where it can show each country with their medals won in different years or overall years. To achieve this I considered **Medals** in columns section and excluded the **Not Achieved** any medals by Athletes. In row, I selected **Team** and **CNT(Athlete_Events)**. Also used **Year** as filter to get the medals in respective years by the Teams. The Pre-Attentive attribute that we can identify is length .

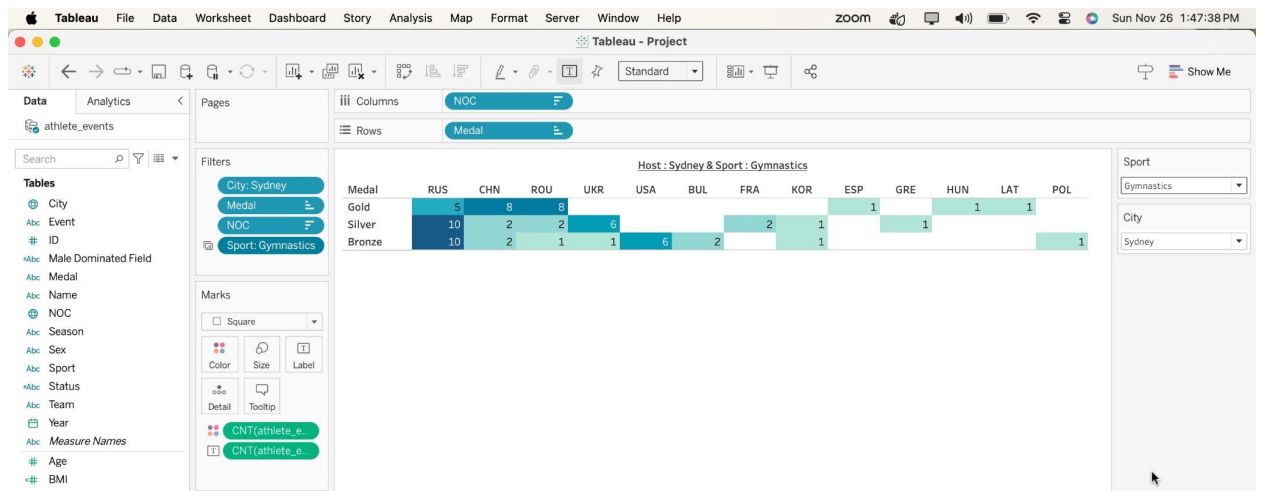
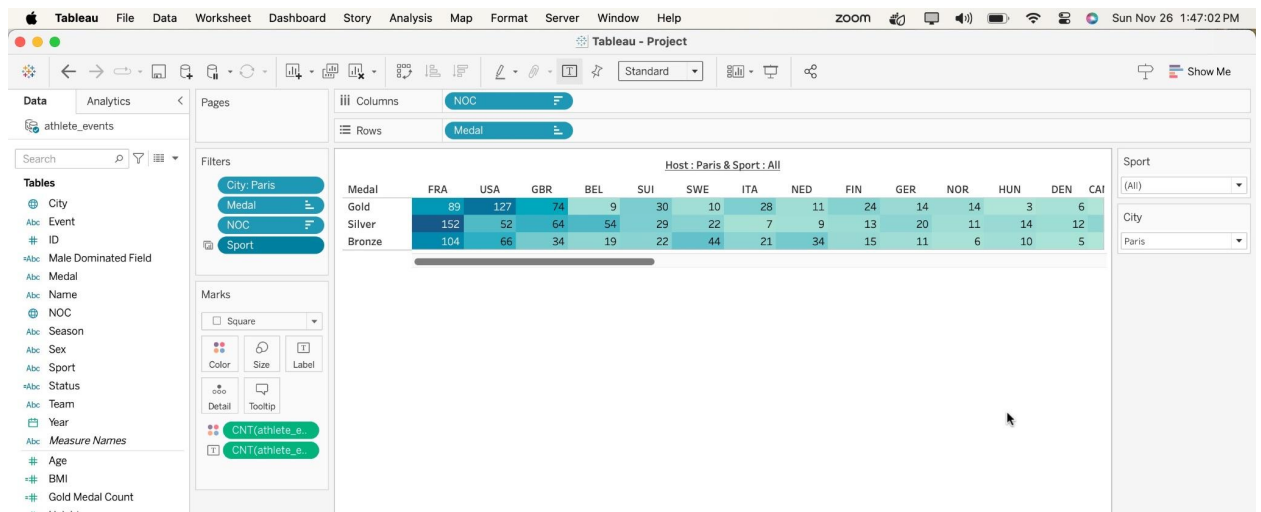
- **Length:** This defines the total number of medals achieved by each team in different years.





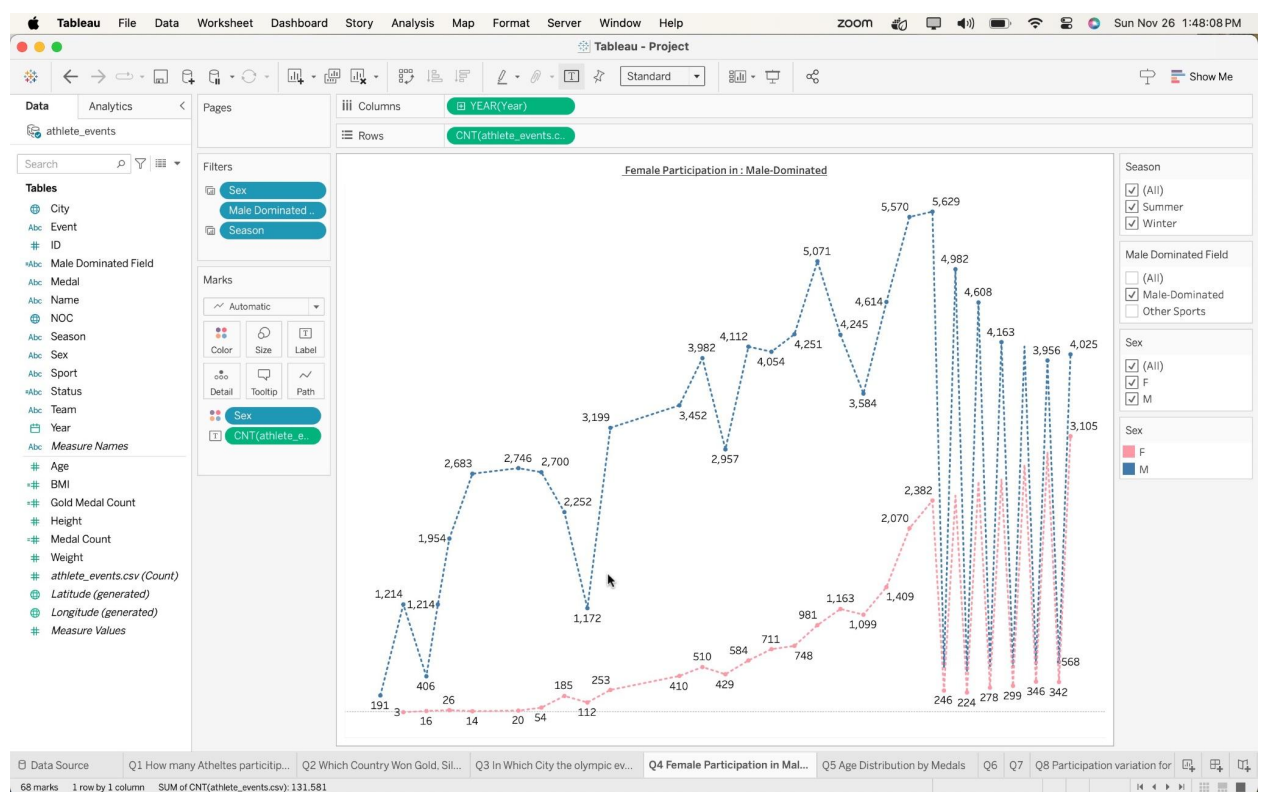
3. For each host country, what is the count of different medals achieved by the different teams.

In this visualization, we are trying to show the medals achieved by the different teams based on Olympic events hosted by countries in their major cities. This visualization is shown in Table format having 3 rows, Gold, Silver and Bronze (Excluded Not Achieved Athletes). And the Columns of the country got medals. Using the **City** as a filter, we can see how many medals were won while the Olympic event is hosted in the city.

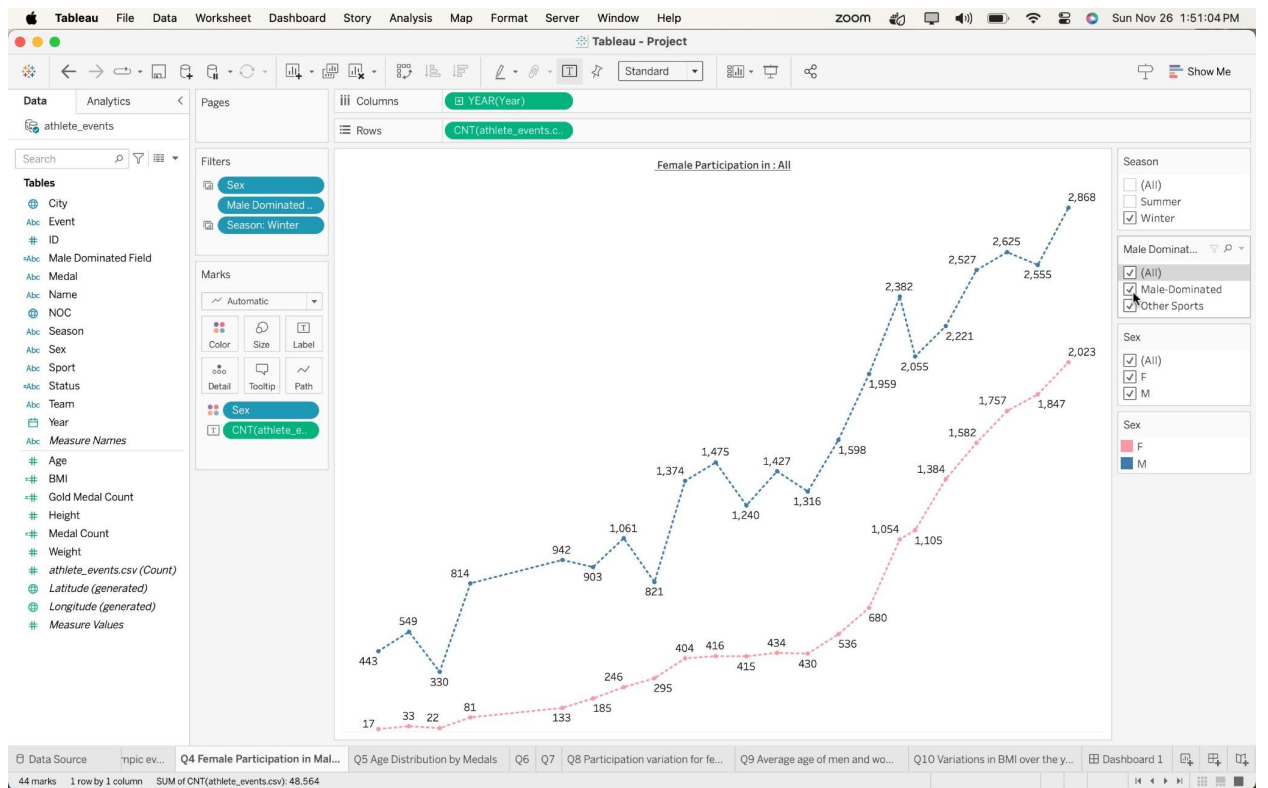
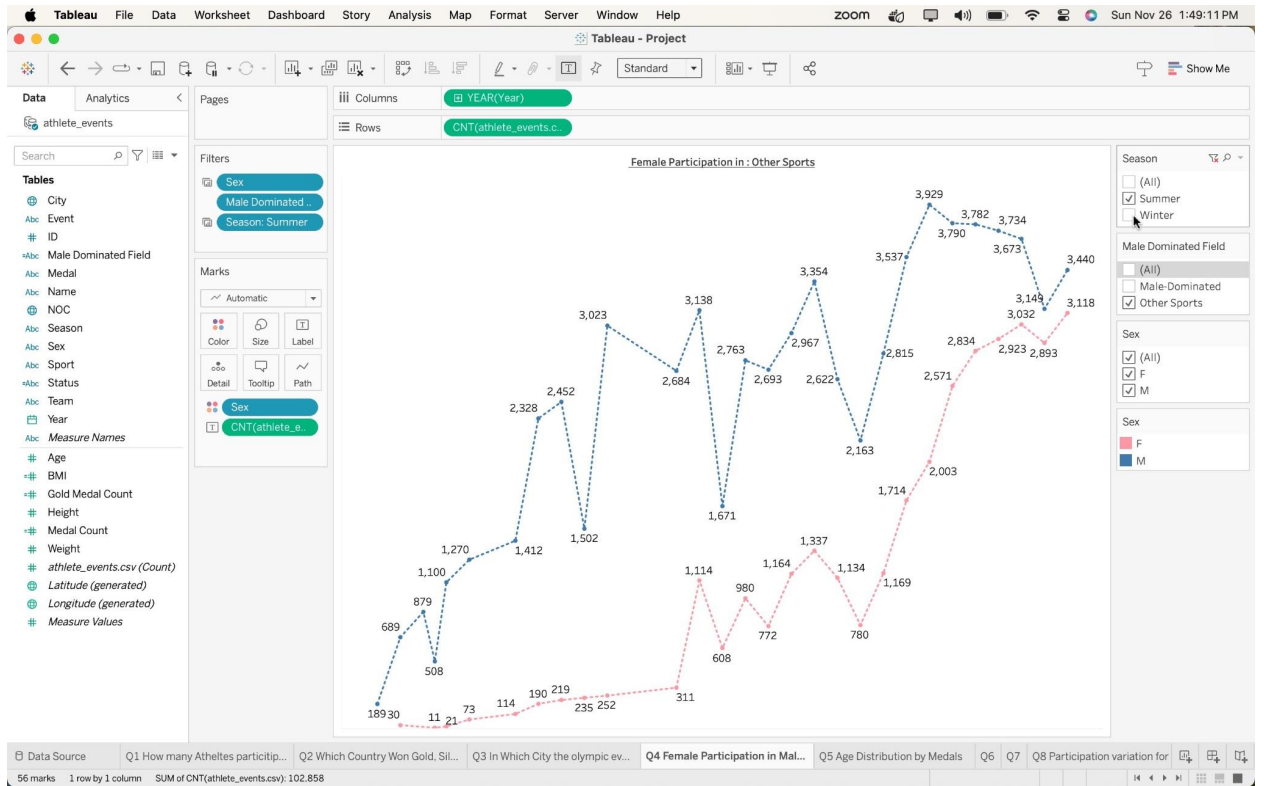


4. Has there been an increase in female participation over the years particularly in often male dominated sports?

This visualization tracks the rise of female participation in the Olympics, specifically in traditionally male-dominated sports. Using a two-column table, it showcases yearly data, distinguishing between male and female participants. A color scheme or shape helps differentiate genders, while a filter isolates male-dominated sports for focused insights.

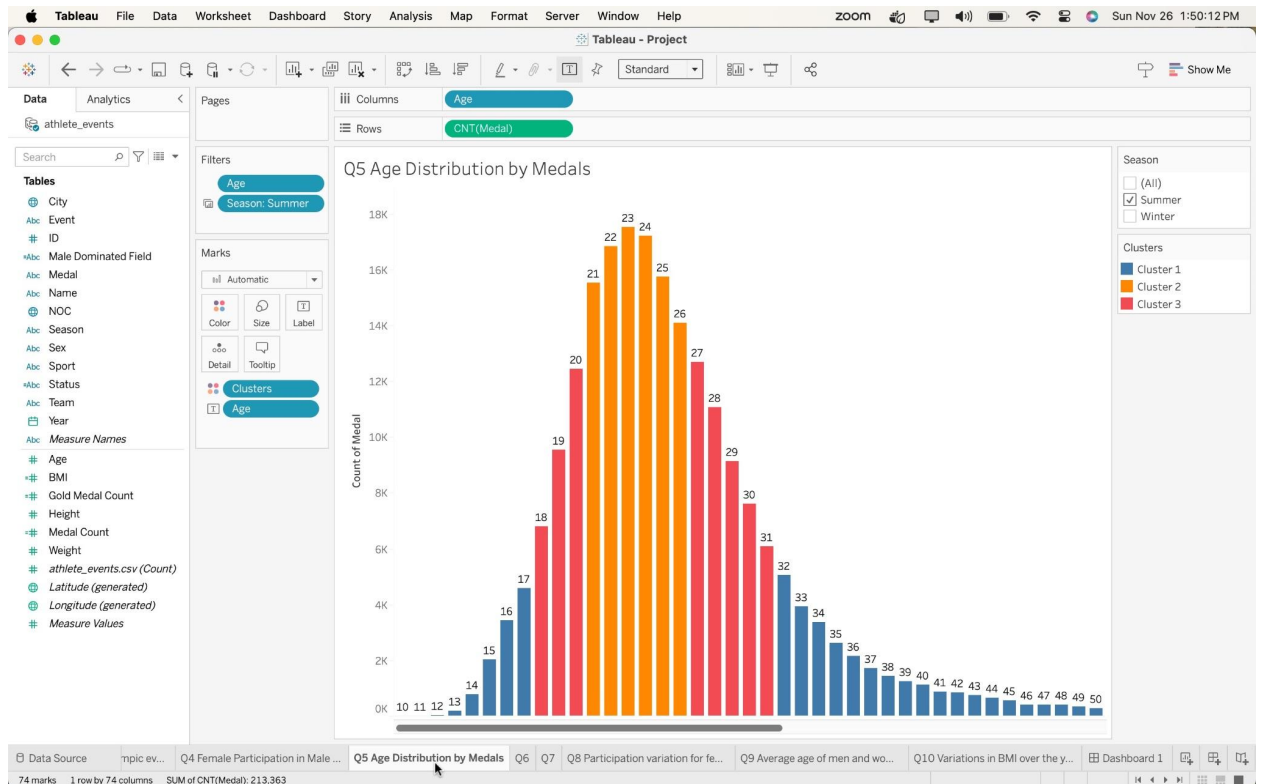


The goal is to provide a comprehensive view of the evolving landscape of female participation in the Olympics. The trend of count of athlete_events.csv for Year. Color shows details about Sex. The marks are labeled by count of athlete_events.csv. The data is filtered on Male Dominated Field and Season. The Male Dominated Field filter keeps Male-Dominated. The Season filter keeps Winter. The view is filtered on Sex, which keeps multiple members.



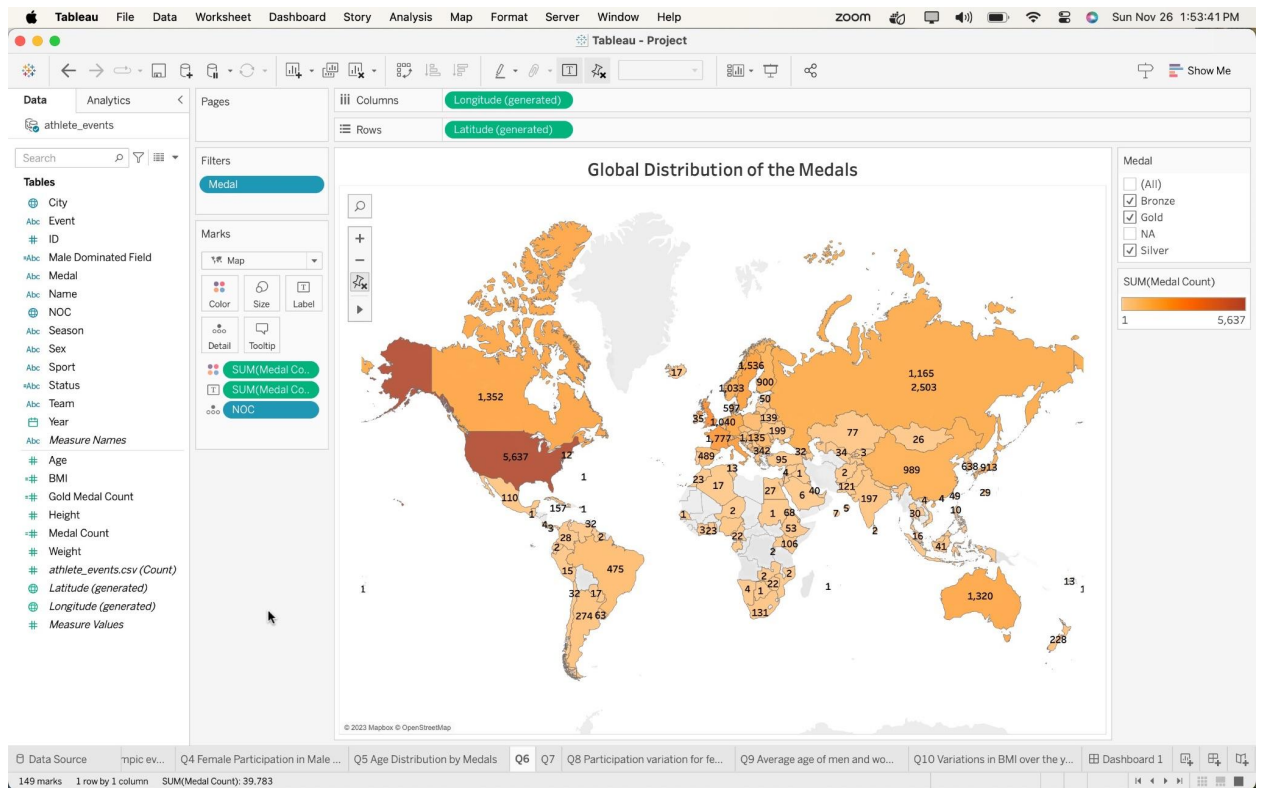
5. What is the distribution of the age of gold medalists in either the summer or winter seasons?

This visualization illustrates the age distribution of gold medalists, differentiating between summer and winter seasons. Using a two-column structure, it displays age groups with distinct visual cues. A filter allows users to focus on either season



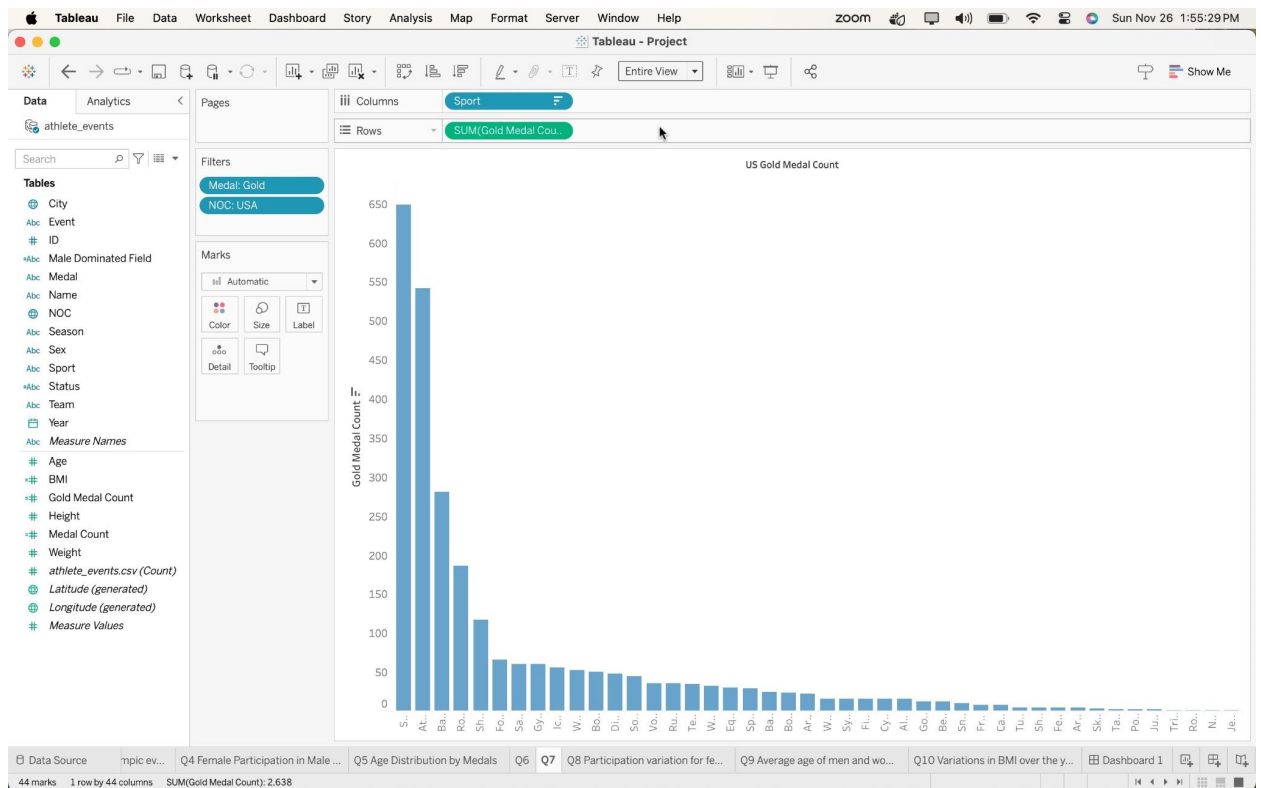
6. How evenly are the acquired medals distributed on a global scale?

This visualization showcases the global distribution of Olympic medals won over the 120-year history of the Games and analyzes whether there has been a fair spread or distribution.



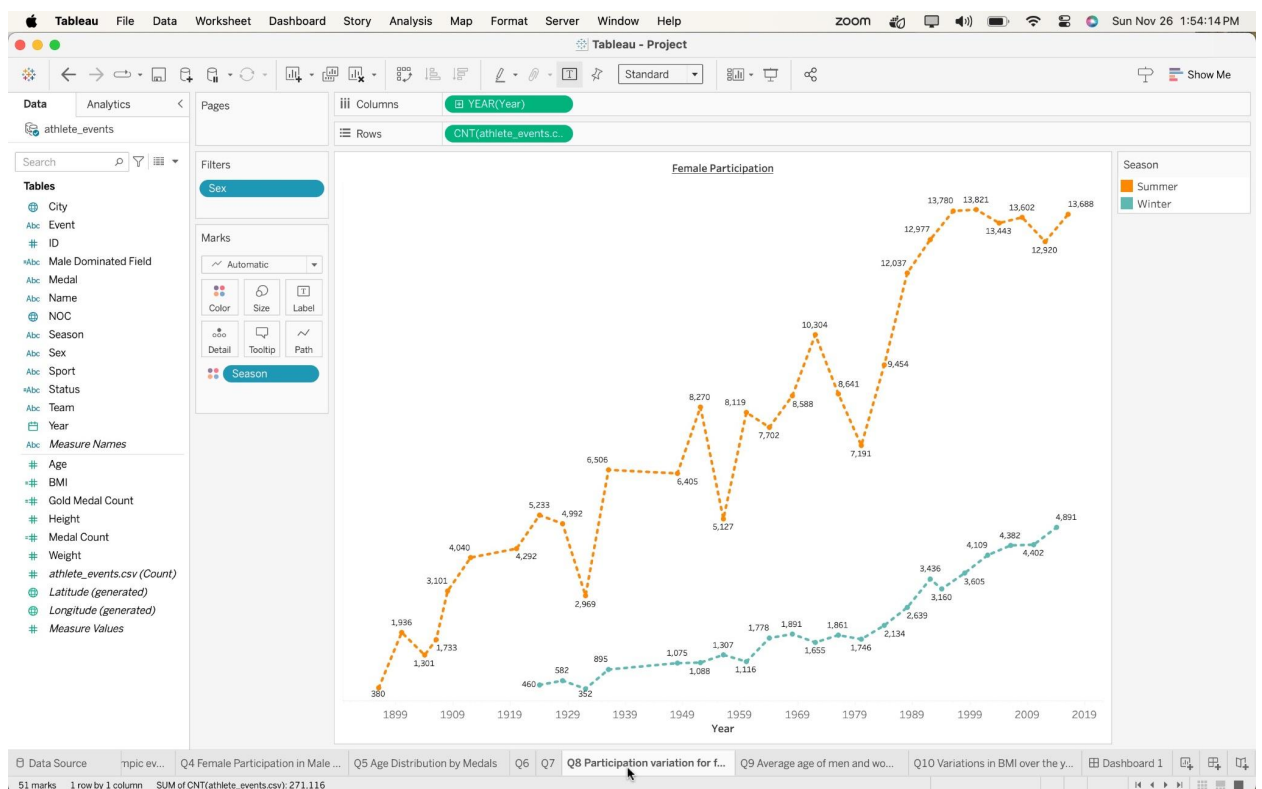
7. In which Olympic sport or discipline has the United States achieved the fewest gold medals over the course of the 120-year history of the Games?

Sum of Gold Medal Count for each Sport. The marks are labeled by sum of Gold Medal Count. The data is filtered on Medal and NOC. The Medal filter keeps Gold. The NOC filter keeps the USA.



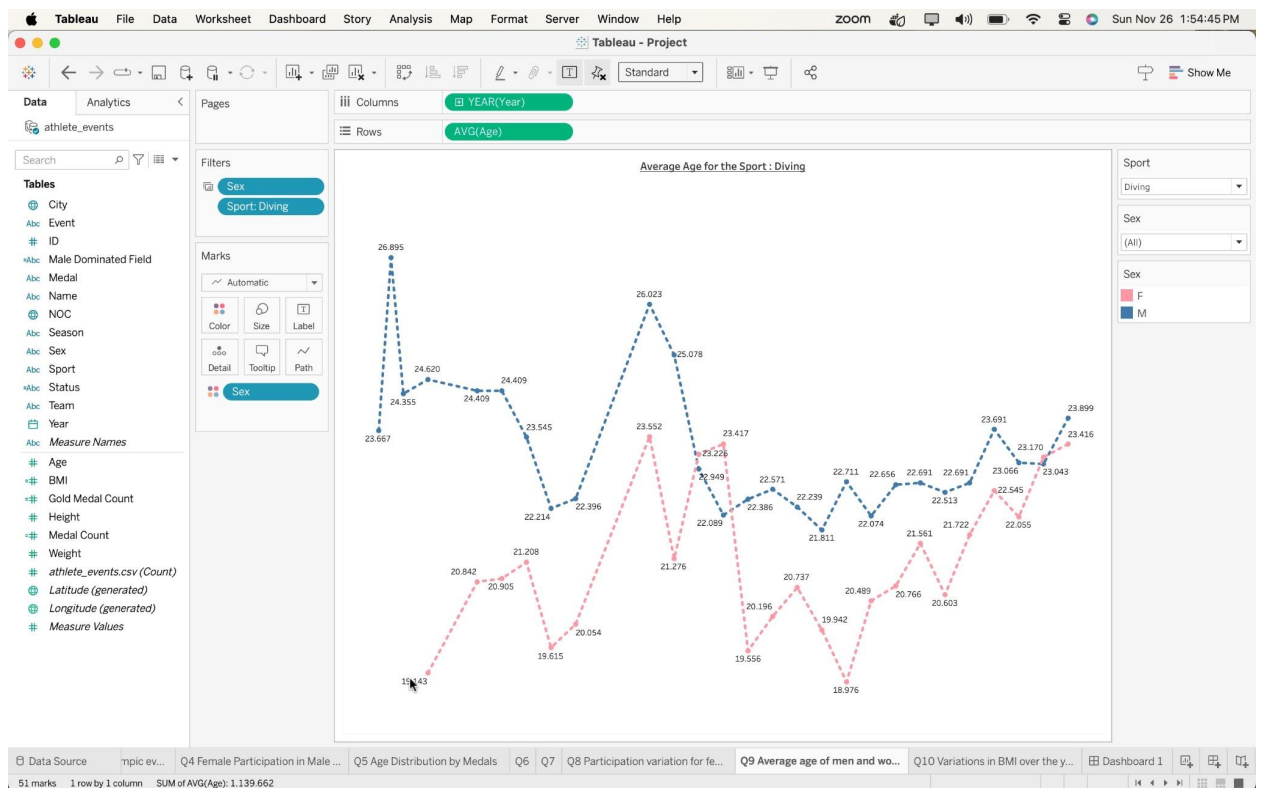
8. What is the variations for female participation for the different seasons over the years

The chart shows a growing female participation rates in the olympic season with more women involved in the summer sports compared to the winter sports.



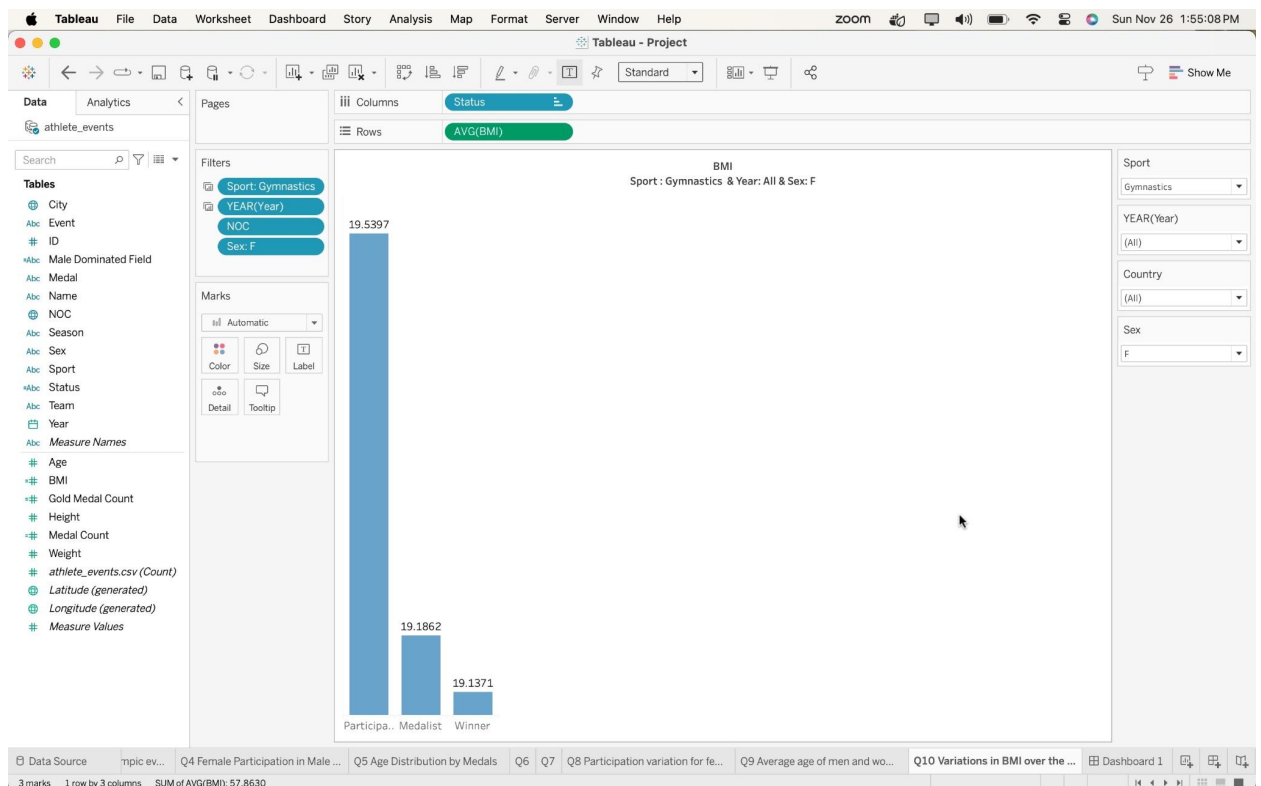
9. For selected teams(countries), what is the average age of women and men participating in the Olympics over the years

- a. The line chart below was chosen to provide a clean overview of how the average age of participants has changed over the years. The audience are able to compare different sexes, countries, and also compare how historic events such as the world wars may have affected the average age of participants.



10. For the different sports, how has the average BMI of winners changed over the past years compared to non-winning participants?

This chart shows how the average BMI of participants have changed over the years for the different sports. Allowing the audience to filter by sports, country, and year. The audience are also able to compare the BMI of the respective winners, other medalists and all other participants. We excluded the sport who are not having the height and weight values. We have not removed the data because few athletes have won the medals. For that reason we haven't removed it because all other questions may have effects



Section 6: Interactivity

Tableau users enjoy a range of customization options to tailor their dashboards. Parameters, for instance, play a pivotal role in seamlessly transitioning between metrics, managing sorting preferences, and executing complex computations. By adjusting parameters like numerical values or computed fields, users can meet their specific requirements, influencing various aspects of worksheet operations, filters, and reference lines. The utilization of parameters, calculated fields, and filters allows for the integration of interactive controls in visualizations, providing a versatile approach to data analysis and presentation.

Seasons:

The "Season" attribute distinguishes between two distinct seasons in the visualization, aiding the identification of events based on the respective season for addressing question 1. The stacked bar chart within the visualization effectively portrays both the summer and winter seasons, visually represented by two distinct colors. This visual representation provides clarity and allows users to discern the occurrence of events in each season. The utilization of a stacked bar chart enhances the visual appeal and facilitates a comprehensive understanding of the distribution of events across summer and winter seasons.



Year:

The year filter serves as a crucial tool for pinpointing the teams or countries that secured medals in specific Olympic years. This filter is conveniently presented in a single dropdown list, allowing users to navigate through various years effortlessly. By leveraging this filter, users can gain insights into the

achievements of different countries across different Olympic events in respective years. It provides a comprehensive view of how teams demonstrated their prowess in various events, showcasing their talents and accomplishments over the years. The user-friendly dropdown feature enhances the accessibility of information related to medal-winning performances, fostering a deeper understanding of each country's Olympic journey throughout the years.

Year

(All)

▼

Enter search text

(All)

1896

1900

1904

1906

1908

1912

1920

1924

1928

1932

1936

1948

1952

1956

1960

1964

1968

1972

1976

1980

1984

1988

City:

The city filter serves as a valuable tool for pinpointing the teams or countries that secured medals in Olympic events hosted in a specific city. This filter is conveniently presented in a single dropdown list, allowing users to select the desired city of interest. By utilizing this filter, users can gain insights into the achievements and medal standings of various countries in the context of the chosen host city. The interactive nature of the dropdown list enhances the user experience, enabling them to explore and analyze Olympic data based on specific geographic locations. This feature proves particularly useful for understanding the performance and success of nations in different Olympic venues.



The image shows a web interface for a city filter. At the top, there is a label "City" next to a small red 'x' icon and a downward arrow. Below this is a dropdown menu with "Paris" selected. Underneath the dropdown is a search bar with the placeholder text "Enter search text". Below the search bar is a list of cities. The cities listed are: (All), Albertville, Amsterdam, Antwerpen, Athina, Atlanta, Barcelona, Beijing, Berlin, Calgary, Chamonix, Cortina d'Ampezzo, Garmisch-Partenkirchen, Grenoble, Helsinki, Innsbruck, Lake Placid, Lillehammer, London, Los Angeles, Melbourne, Mexico City, Montreal, Moskva, Munich, Nagano, Oslo, Paris (highlighted), Rio de Janeiro, and Roma.

City
(All)
Albertville
Amsterdam
Antwerpen
Athina
Atlanta
Barcelona
Beijing
Berlin
Calgary
Chamonix
Cortina d'Ampezzo
Garmisch-Partenkirchen
Grenoble
Helsinki
Innsbruck
Lake Placid
Lillehammer
London
Los Angeles
Melbourne
Mexico City
Montreal
Moskva
Munich
Nagano
Oslo
Paris
Rio de Janeiro
Roma

Country :

Interactive filters such as country are helpful to enable users to track the performance of a particular country of interest or compare trends with other countries. For instance users could compare two developing countries or economically different countries to analyze how the economic difference may have been reflected in sports.

Country

(All)

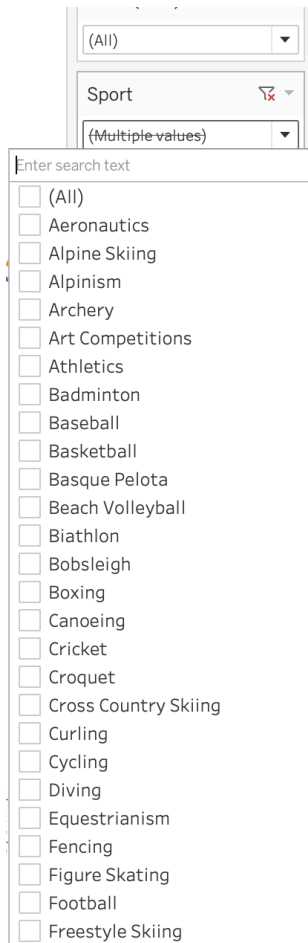
Enter search text

- ☒ (All)
- ☒ AFG
- ☒ AHO
- ☒ ALB
- ☒ ALG
- ☒ AND
- ☒ ANG
- ☒ ANT
- ☒ ANZ
- ☒ ARG
- ☒ ARM
- ☒ ARU
- ☒ ASA
- ☒ AUS
- ☒ AUT
- ☒ AZE
- ☒ BAH
- ☒ BAN
- ☒ BAR
- ☒ BDI
- ☒ BEL
- ☒ BEN
- ☒ BER
- ☒ BHU
- ☒ BIH
- ☒ BIZ
- ☒ BLR

Sports:

For most fans, the choice of sports they follow are influenced by personal interest. Some of such differences are also recorded across regions with some sports fans following soccer while other may prefer to view the data related to basketball. Accordingly, visualizations depicting average BMI and average age of participants and winners also allow audiences to filter by selecting their preferred sports. This interactive filter is designed for use not only by the fans, but also prospective olympic participants, as such, the filter allows the user to select more than one sport,

and to see how these values differ across. This will enable a prospective athlete hoping to participate in more than one event to consider these values carefully during the training preparation phase.



The image shows a user interface for filtering sports. At the top, there is a dropdown menu currently set to "(All)". Below it is a section labeled "Sport" with a red 'x' icon and a dropdown arrow. Underneath the "Sport" label is another dropdown menu currently set to "(Multiple values)". Below these is a search bar with the placeholder text "Enter search text". Below the search bar is a list of sports, each preceded by an unchecked checkbox. The sports listed are: (All), Aeronautics, Alpine Skiing, Alpinism, Archery, Art Competitions, Athletics, Badminton, Baseball, Basketball, Basque Pelota, Beach Volleyball, Biathlon, Bobsleigh, Boxing, Canoeing, Cricket, Croquet, Cross Country Skiing, Curling, Cycling, Diving, Equestrianism, Fencing, Figure Skating, Football, and Freestyle Skiing.

Sport
☐ (All)
☐ Aeronautics
☐ Alpine Skiing
☐ Alpinism
☐ Archery
☐ Art Competitions
☐ Athletics
☐ Badminton
☐ Baseball
☐ Basketball
☐ Basque Pelota
☐ Beach Volleyball
☐ Biathlon
☐ Bobsleigh
☐ Boxing
☐ Canoeing
☐ Cricket
☐ Croquet
☐ Cross Country Skiing
☐ Curling
☐ Cycling
☐ Diving
☐ Equestrianism
☐ Fencing
☐ Figure Skating
☐ Football
☐ Freestyle Skiing

Gender:

The gender filter primarily provides an opportunity for users of the dashboard who maybe interested in comparing the different values to understand how the visualizations change with different sexes of the participants in the olympic games.

Country

(All) ▼

Gender Filter

(All) ▼

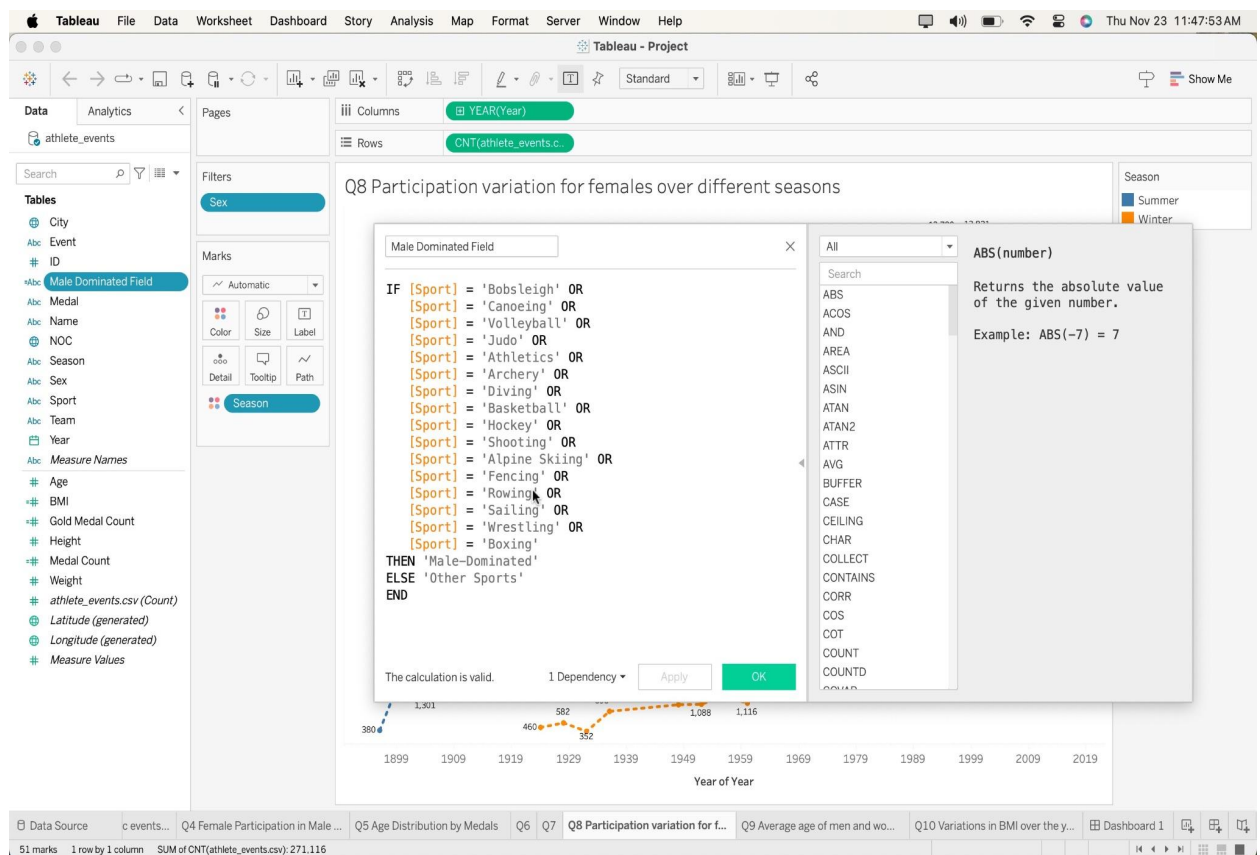
(All)

Female

Male

Male Dominated Field Parameter:

This sports filter adds another layer of specificity and interactivity to our dashboard. It introduces a layer of differentiation between sports that have been predominantly male centric over those that have not in the 120 year history of the olympics. It allows us to filter the visualizations for male dominated sports vs other sports and analyze trends of either genders based on their seasons of competition.



Male Dominat...   ▼

- ☐ (All)
- ☒ Male-Dominated
- ☐ Other Sports

Season

- ☐ (All)
- ☒ Summer
- ☐ Winter

References

Dataset Link: Depinto, M. G. (2018, September 23). Let's discover more about the Olympic Games! Kaggle.

<https://www.kaggle.com/code/marcogdepinto/let-s-discover-more-about-the-olympic-games/notebook#7.-Medals-per-country>

Mural 23 Link:

<https://app.mural.co/t/datavisualizationfinalprojee5159/m/datavisualizationfinalprojee5159/1699563637854/1cc71ea8316fec9084fe918b1eabef52bdd8549e?sender=uc73c99f02de719af19555441>

Published Link:

https://public.tableau.com/views/IFT_533_Final_Project_Olympic_History_by_Numbers/Dashboard1?:language=en-US&publish=yes&:display_count=n&:origin=viz_share_link